A-level

ECONOMICS

Paper 1 Markets and market failure

Predicted Paper Set A — Moderate

MARK SCHEME and MODEL ANSWERS

Total paper mark: 80 | Section A: 40 marks | Section B: 40 marks

General marking guidance

- Examiners reward responses for what candidates have demonstrated, not penalised for omissions.
- Where a response falls between two levels, the mark awarded should reflect the quality of the analysis and evaluation, not just the number of points made.
- Diagrams must be accurate and fully labelled to score full marks. A diagram that is described clearly in words may earn partial credit.
- Indicative content is not exhaustive. Candidates may offer alternative relevant arguments and examples which should be credited on merit.
- For evaluation questions, credit the quality of judgement — a clear, reasoned conclusion supported by analysis is essential for the top level.

Assessment Objectives (AOs)

AO1 Knowledge and understanding of economic terms, concepts, theories and models.

AO2 Application of economic knowledge and understanding to various contexts.

AO3 Analysis of economic issues, showing understanding of their impact on economic agents.

AO4 Evaluation of economic arguments and use of qualitative and quantitative evidence to support informed judgements.

Section A — Data Response

The UK Soft Drinks Industry Levy

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0 1	*Calculate the percentage change in sugary drink consumption, 2018–2024.*	**2 marks**

Indicative content

Formula: Percentage change = ((New − Old) / Old) × 100

Working: ((52.7 − 78.4) / 78.4) × 100 = (−25.7 / 78.4) × 100 = −32.78% ≈ −32.8%

Answer: A **decrease of 32.8%** (accept 32.7% or 32.8% to 1 d.p.).

Mark allocation

- **1 mark** — correct numerical answer of 32.8% (accept 32.7%).
- **1 mark** — correctly identifying the change as a decrease (or negative value correctly signed).

Award 2 marks for a fully correct answer of − 32.8% (a decrease) with or without working shown. Award 1 mark for correct working but wrong final figure, or for the correct figure without the direction stated.

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0 2	*Diagram and explanation: effect of the sugar tax on the market for high-sugar soft drinks.*	**4 marks**

Indicative diagram

A standard supply-and-demand diagram with:

- Price on the vertical axis, Quantity on the horizontal axis.
- Downward-sloping demand curve (D).
- Upward-sloping supply curve (S).
- Supply curve shifts vertically upwards (leftwards) by the amount of the tax to S + tax.
- New equilibrium: price rises from P_1 to P_2; quantity falls from Q_1 to Q_2.
- Tax revenue shaded as the rectangle between the two supply curves up to Q_2.

Indicative written explanation

The Soft Drinks Industry Levy is an indirect tax imposed on producers. It raises the marginal cost of producing each litre of sugary drink by 18p or 24p depending on the sugar content. This causes the supply curve to shift vertically upwards by the value of the tax. At the new equilibrium, the market price rises from P_1 to P_2, and the equilibrium quantity falls from Q_1 to Q_2. Producers pass some of the tax on to consumers (consumer incidence) while absorbing some themselves (producer incidence); the relative shares depend on the price elasticities of demand and supply.

Mark allocation (4 marks)

- **1 mark** — correctly labelled axes (Price and Quantity).
- **1 mark** — correctly drawn initial S and D curves showing equilibrium P_1Q_1.
- **1 mark** — correct upward/leftward shift of the supply curve showing new equilibrium P_2Q_2.
- **1 mark** — written explanation of the mechanism (tax → higher production cost → reduced supply → higher price and lower quantity).

A clear verbal description of the diagram may earn up to 3 marks where a visual diagram is absent but the mechanism is fully explained.

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0 3	*Analyse why the sugar tax may be considered an example of government intervention to correct market failure.*	**9 marks**

Required economic content

- Definition of market failure — the free market fails to allocate resources efficiently, producing too much or too little of certain goods relative to the socially optimal level.
- Negative externality of consumption — private consumers of sugary drinks do not bear the full social cost of their decision (NHS costs, productivity losses, external third-party costs).
- At the free-market equilibrium, marginal private benefit (MPB) exceeds marginal social benefit (MSB); sugary drinks are over-consumed relative to the socially optimal level.
- Welfare loss triangle represents the deadweight loss between the free-market quantity Q_1 and the socially optimal quantity Q*.

Required diagram

Negative-consumption-externality diagram showing:

- MPC = MSC (no production externality assumed).
- MPB above MSB, with vertical gap equal to the marginal external cost.
- Free-market equilibrium at Q_1 where MPB = MSC.
- Socially optimal equilibrium at Q* where MSB = MSC.
- Welfare loss triangle shaded between Q* and Q_1.
- Tax shifts MPB downwards (or shifts supply up) to align private behaviour with the social optimum.

Application using the extracts

- **Extract A:** The NHS incurs approximately £6 billion per year from excess sugar consumption — direct evidence of the external cost.
- **Extract B:** SDIL revenue rose from £240m (2018) to £415m (2024) while consumption fell from 78.4 to 52.7 litres per capita — shows the tax has reduced consumption towards the social optimum.
- **Extract A:** Reformulated drinks accounted for over 50% of sales by volume in 2023 — producer behaviour has also shifted, reducing sugar content without needing to pass the full tax onto consumers.

Chain of reasoning expected

Tax → raises producers' costs → shifts supply curve left → raises market price → reduces consumer demand (law of demand) → lowers equilibrium quantity closer to the socially optimal level Q* → reduces the external costs imposed on the NHS and wider society → improves allocative efficiency and social welfare. The tax also generates revenue that can be hypothecated to health initiatives, reinforcing the corrective effect.

Mark scheme — levels

Level	Marks	Descriptor
Level 3	7–9	Analysis is detailed and logical, with a fully labelled and accurate externality diagram. The response makes clear use of both extracts and develops at least one full chain of reasoning linking the tax to the correction of the market failure.
Level 2	4–6	Sound analysis with a reasonable diagram. Some reference to the extracts. Chain of reasoning is present but may be partial or contain minor errors.
Level 1	1–3	Limited analysis. Diagram may be missing, incomplete or contain significant errors. Little application to the extracts.

0 4	*Evaluate the view that indirect taxes are the most effective way to reduce consumption of demerit goods.*	**25 marks**

Arguments supporting indirect taxation (FOR)

- **Effective price signal:** taxes raise the market price of demerit goods, reducing demand. Extract B shows UK sugary drink consumption fell by 32.8% between 2018 and 2024, coinciding with rising SDIL revenue.
- **Internalises the externality:** taxes bring the marginal private cost closer to the marginal social cost, improving allocative efficiency.
- **Revenue generation:** SDIL raised £415m in 2024 which can be hypothecated to fund health campaigns, school food programmes or the NHS — partially offsetting the £6 billion annual cost.
- **Drives reformulation:** Extract A notes that reformulated drinks accounted for over 50% of sales by volume in 2023. Producers reduce harmful content to avoid the tax — a supply-side benefit that persists regardless of elasticity of demand.
- **Institutional endorsement:** the IFS described the SDIL as 'one of the most successful examples of behavioural government intervention'.

Arguments against / evaluation (AGAINST)

- **Regressive incidence:** Extract C shows the lowest income decile spent 0.7% of disposable income on SDIL-affected products versus 0.2% for the highest decile. Indirect taxes disproportionately affect lower-income households, worsening inequality.
- **Inelastic demand for addictive/habitual goods:** where demand is price inelastic (e.g. tobacco, alcohol for heavy users), taxes have a small effect on consumption but a large effect on revenue and welfare — a weak corrective but strong fiscal tool.
- **Substitution effects:** consumers may switch to untaxed or less visible substitutes. The SDIL does not (yet) cover milkshakes or sugary coffees, potentially displacing rather than reducing sugar intake.
- **Black markets and cross-border shopping:** higher rates can induce smuggling or cross-border purchasing, eroding the tax base without reducing consumption.
- **Information failure persists:** a tax alone does not educate consumers about long-run health consequences; behavioural change may be temporary.

Alternative policies (for comparative evaluation)

- Advertising restrictions — particularly effective for children where bounded rationality is severe.
- Mandatory front-of-pack labelling and reformulation targets — tackle information failure directly.
- Education and public-health campaigns — long-run behavioural change, but slow and often fail to reach at-risk groups.
- Minimum unit pricing — prevents retailers absorbing the tax, and directly targets the cheapest, highest-strength products (used in Scotland for alcohol).
- Subsidies for healthy alternatives — shift relative prices without regressive impact.
- Nudge interventions (default options, social-norm messaging) — low cost, non-coercive, but often weak in isolation.

Expected judgement and conclusion

A top-band answer should reach a reasoned judgement. The SDIL case study suggests indirect taxation is highly effective but works most powerfully alongside complementary interventions — particularly reformulation targets and advertising restrictions. The WHO (2023) and LSE evidence cited in the extracts supports a 'policy mix' conclusion: taxation is a necessary but rarely sufficient instrument. Therefore the claim that indirect taxes are the single 'most effective' tool is only partly justified — effectiveness depends critically on elasticity, substitutability, complementary policy, and the distributional design of the tax (e.g. tiered thresholds that encourage reformulation).

Mark scheme — levels

Level	Marks	Descriptor
Level 5	21–25	Sophisticated analysis and evaluation. Sustained chains of reasoning; accurate diagram(s); substantial use of both extracts and own knowledge; clear, well-supported judgement.
Level 4	16–20	Good analysis and evaluation. Diagram present and mostly accurate. Two or more developed evaluative points. Supported judgement, possibly less nuanced.
Level 3	11–15	Sound analysis with some evaluation. Use of extracts is present but may be descriptive. Judgement may be asserted rather than reasoned.
Level 2	6–10	Some relevant analysis. Limited evaluation. Weak or partial use of extracts.
Level 1	1–5	Limited analysis; evaluation absent or generic. Little or no use of extracts.

Section B — Essay

Public goods, externalities and the free market

0 5	*Explain how the existence of negative externalities and public goods can lead to market failure.*	**15 marks**

Required definitions

- **Market failure:** a situation in which the free market fails to allocate scarce resources to their most valued use — the socially optimal quantity is not produced or consumed.
- **Negative externality:** a spillover cost imposed on a third party not involved in the economic transaction. The marginal social cost (MSC) exceeds the marginal private cost (MPC), or marginal social benefit is below marginal private benefit.
- **Public good:** a good that is both non-rival (one person's consumption does not reduce availability to others) and non-excludable (consumers cannot be prevented from benefiting).

Part A — Negative externalities

In the presence of a negative externality in production (e.g. pollution from a factory), MSC exceeds MPC. The free market equates supply (based on MPC) with demand, yielding equilibrium quantity Q_1 at price P_1. The socially optimal level, where MSC intersects demand, is at $Q^* < Q_1$. The difference $Q_1 - Q^*$ represents overproduction. The triangular area between MSC and MPC from Q^* to Q_1 represents the welfare loss (deadweight loss) caused by the externality.

In the presence of a negative externality in consumption (e.g. passive smoking, obesity-related NHS costs), MSB lies below MPB. The free market at Q_1 overconsumes relative to Q^*, again generating a welfare loss. Because individual producers and consumers do not pay for or receive the external effects, they have no incentive to reduce output or consumption to the socially optimal level — the price mechanism fails to transmit the full social cost or benefit.

Required diagram — negative externality

- Axes: Price/Cost on y-axis, Quantity on x-axis.
- MPC (upward-sloping), MSC (above and parallel to or diverging from MPC).
- Demand = MPB curve (downward-sloping).
- Free-market equilibrium Q_1 at MPC = D; social optimum Q^* at MSC = D.
- Welfare loss triangle shaded between Q^* and Q_1.

Part B — Public goods

Public goods suffer from the free-rider problem: because consumers cannot be excluded from benefiting (non-excludability), they have no incentive to pay voluntarily. If private firms attempt to supply the good, they cannot recover their costs because non-payers cannot be denied the benefit. Additionally, non-rivalry means the marginal cost of an extra user is zero, so efficient pricing would mean a zero price — but at zero revenue no private firm will supply the good. The result is a missing market: the good is either not supplied at all or is chronically under-supplied. Classic examples include national defence, street lighting, flood defences and lighthouses, none of which could be provided efficiently by the private sector without state intervention.

Expected content for top-band responses

- Clear definitions and correct use of MPC, MSC, MPB, MSB terminology.
- At least one fully labelled diagram (externality, and optionally demand for a public good).
- Distinct mechanism for each form of failure — under-provision (public goods) versus overproduction/overconsumption (negative externalities).
- At least one real-world example for each (e.g. CO_2 pollution; national defence).

Mark scheme — levels

Level	Marks	Descriptor
Level 4	13–15	Both concepts fully explained with accurate, well-labelled diagrams and clear chains of reasoning. Appropriate real-world examples. Correct use of terminology throughout.
Level 3	9–12	Both concepts explained soundly, with at least one correct diagram. Some real-world application. Minor inaccuracies or one underdeveloped section.
Level 2	5–8	Some understanding of both concepts, but with notable omissions. Diagram present but partially inaccurate. Limited application.
Level 1	1–4	Descriptive only. One concept may be addressed well while the other is weak or absent. Diagrams missing or incorrect.

0 6	*Evaluate: "Free markets always lead to an efficient allocation of resources."*	**25 marks**

Arguments in favour of the statement (FOR)

- **Price mechanism:** performs signalling, rationing and incentive functions that coordinate millions of decentralised decisions (Hayek).
- **Invisible hand (Adam Smith):** individuals pursuing self-interest inadvertently promote the social good by directing resources to their most valued uses.
- **Allocative efficiency:** in perfectly competitive long-run equilibrium, price equals marginal cost, so society's willingness to pay equals the opportunity cost of production.
- **Productive and dynamic efficiency:** profit incentives drive cost reduction; competition drives innovation and R&D (e.g. UK tech and e-commerce growth, supermarket price competition).
- **Consumer sovereignty:** consumer preferences, expressed through spending, guide producer decisions.
- **Empirical evidence:** centrally planned economies (USSR, Mao-era China) suffered chronic shortages and surpluses; liberalisation since the 1980s has been associated with higher GDP per capita and productivity in most economies.

Arguments against the statement (AGAINST)

- **Public goods:** non-rivalry and non-excludability mean the free rider problem prevents private provision of national defence, street lighting, basic research.
- **Externalities:** markets do not price pollution, passive smoking, or obesity-related health costs. Stern (2007) described climate change as 'the greatest market failure the world has seen'.
- **Merit and demerit goods:** consumers under-consume merit goods (education, vaccines) and over-consume demerit goods (sugar, alcohol, tobacco) due to imperfect information and bounded rationality.
- **Monopoly power:** firms with market power restrict output and raise price above MC, producing allocative inefficiency. UK CMA scrutiny of mergers illustrates ongoing policy concern.
- **Asymmetric information:** Akerlof's 'market for lemons' shows how information asymmetries can cause adverse selection (used cars, insurance, healthcare).
- **Inequity in distribution:** free markets are silent on fairness. An 'efficient' allocation may also be highly unequal — factor markets reward only those with marketable skills or inherited capital.
- **Factor immobility:** structural unemployment in declining regions (e.g. post-industrial UK) persists despite labour market flexibility because of geographical and occupational immobility.

Real-world examples for top-band responses

- 2008 Global Financial Crisis — information asymmetry and moral hazard in mortgage-backed securities; systemic failure requiring massive state intervention.
- UK housing market — planning constraints, inelastic supply and externalities have produced chronic affordability problems.

- Climate change and carbon emissions — price of fossil fuels ignores global externalities; requires carbon pricing or cap-and-trade intervention.
- Pharmaceutical markets — patent monopolies create access problems (insulin, cancer drugs).
- UK energy crisis of 2022 — retail price cap intervention needed to protect households from extreme global price shocks.

Evaluation and judgement

A developed answer should recognise that efficiency is a conditional concept: free markets are efficient only under restrictive assumptions (perfect competition, perfect information, no externalities, well-defined property rights). Where these assumptions fail, as they frequently do, market failure emerges. However, government intervention also incurs costs — regulatory capture, unintended consequences and information problems can produce government failure. The strongest conclusion is therefore nuanced: free markets are often efficient at the micro level, but systematic failures are widespread at the macro level and in specific sectors. A mixed economy, where markets operate within a well-designed regulatory and institutional framework, is empirically the most successful arrangement. The word 'always' in the statement is the crucial failure point — the claim is too strong: efficiency in free markets is common but far from universal.

Mark scheme — levels

Level	Marks	Descriptor
Level 5	21–25	Sophisticated, balanced analysis and evaluation. Uses multiple forms of market failure with accurate diagrams. Includes at least two well-chosen real-world examples. Clear, well-reasoned judgement with sustained evaluative chains.
Level 4	16–20	Good analysis and evaluation. Uses at least two forms of market failure and one real-world example. Supported judgement, possibly less developed than Level 5.
Level 3	11–15	Sound analysis with some evaluation. One or two forms of market failure discussed. Judgement may be asserted rather than reasoned.
Level 2	6–10	Limited analysis. Evaluation weak or superficial. Few examples.
Level 1	1–5	Descriptive only. Little or no evaluation.

Summary of Marks

Question	Topic	AO focus	Marks
0 1	Quantitative skill — percentage change	AO2	2
0 2	Indirect tax — supply and demand diagram	AO1, AO2	4
0 3	Sugar tax correcting market failure	AO1, AO2, AO3	9
0 4	Effectiveness of indirect taxation on demerit goods	AO1–AO4	25
0 5	Negative externalities and public goods	AO1, AO3	15
0 6	Free-market efficiency — evaluation	AO1–AO4	25
TOTAL	**Paper 1: Markets and Market Failure**	**—**	**80**

END OF MARK SCHEME